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Time on Task:
3 hours, 25 minutes

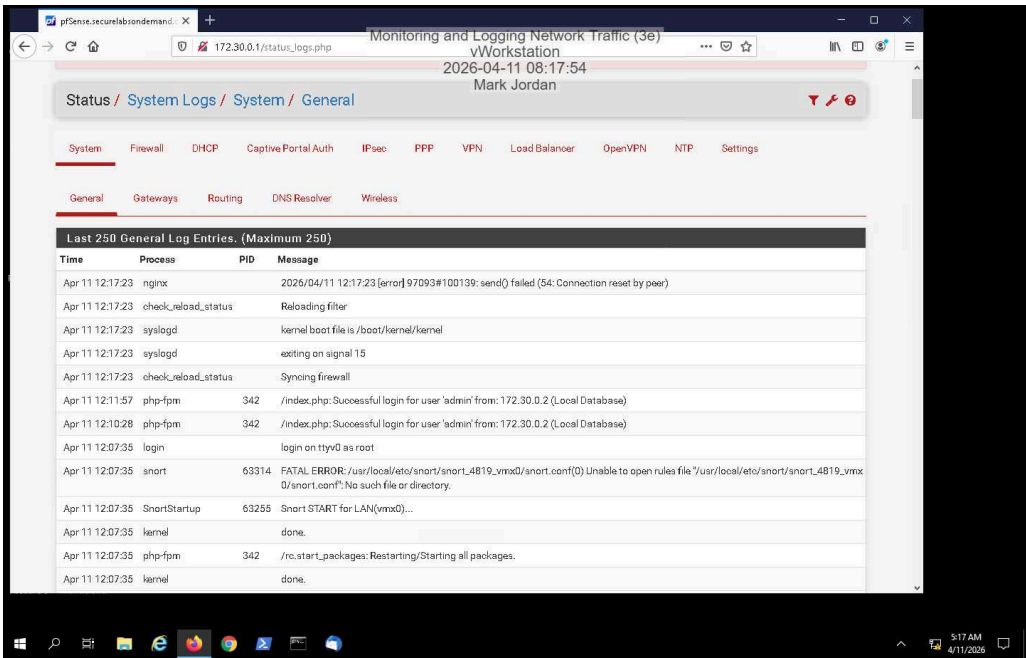
Progress:
100%

Report Generated: Saturday, April 11, 2026 at 11:40 AM

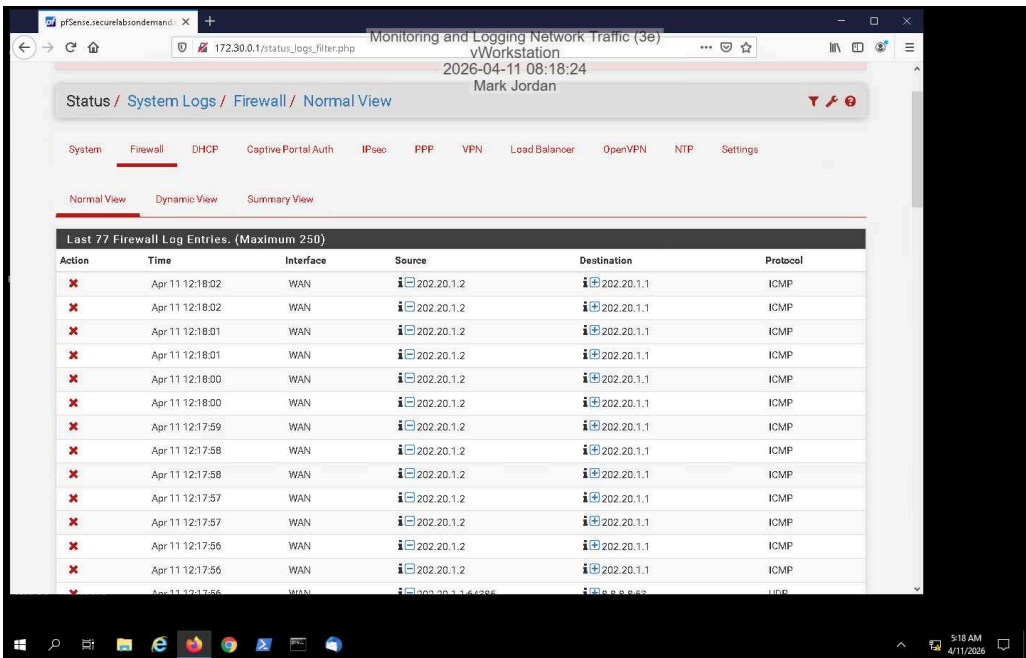
Section 1: Hands-On Demonstration

Part 1: Configure the pfSense Firewall Log

- 13. Make a screen capture showing the system logs.

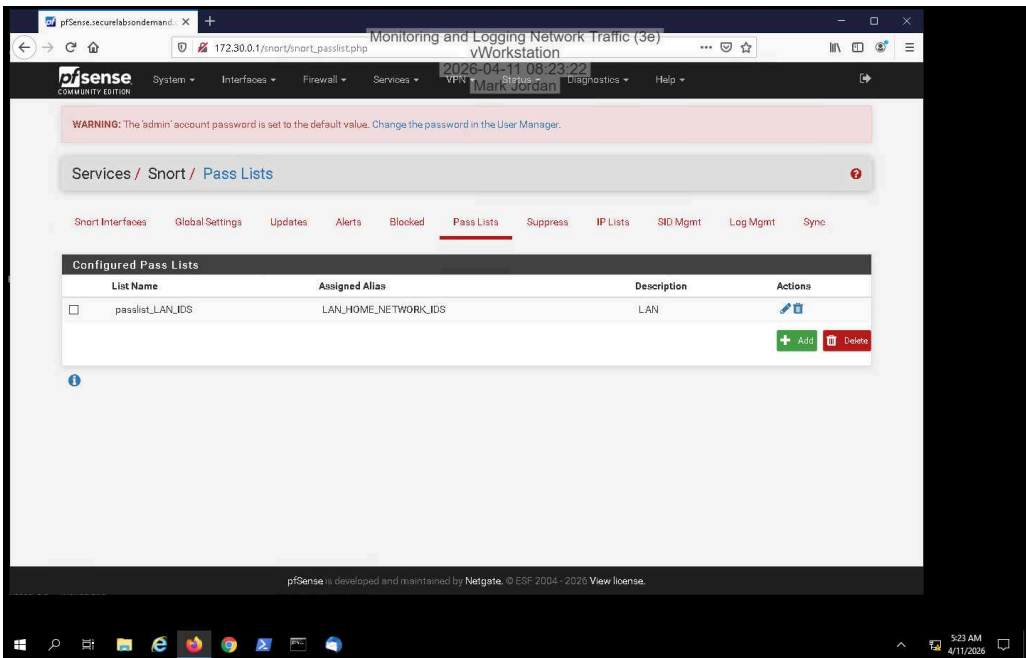


15. Make a screen capture showing the firewall logs.

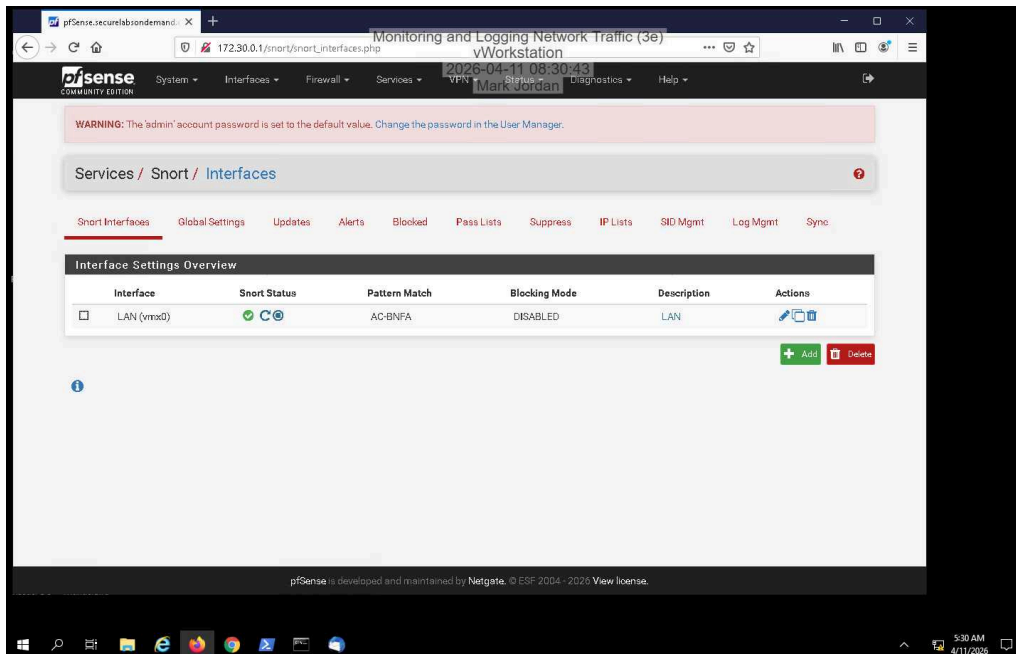


Part 2: Configure a Snort Intrusion Detection System

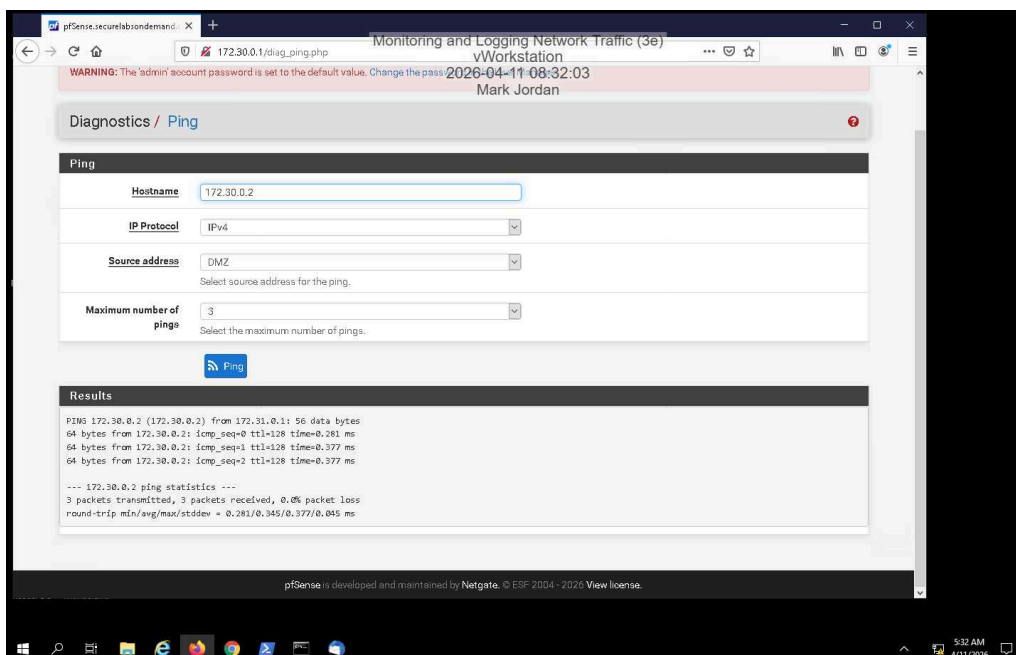
14. Make a screen capture showing the updated Pass Lists page.



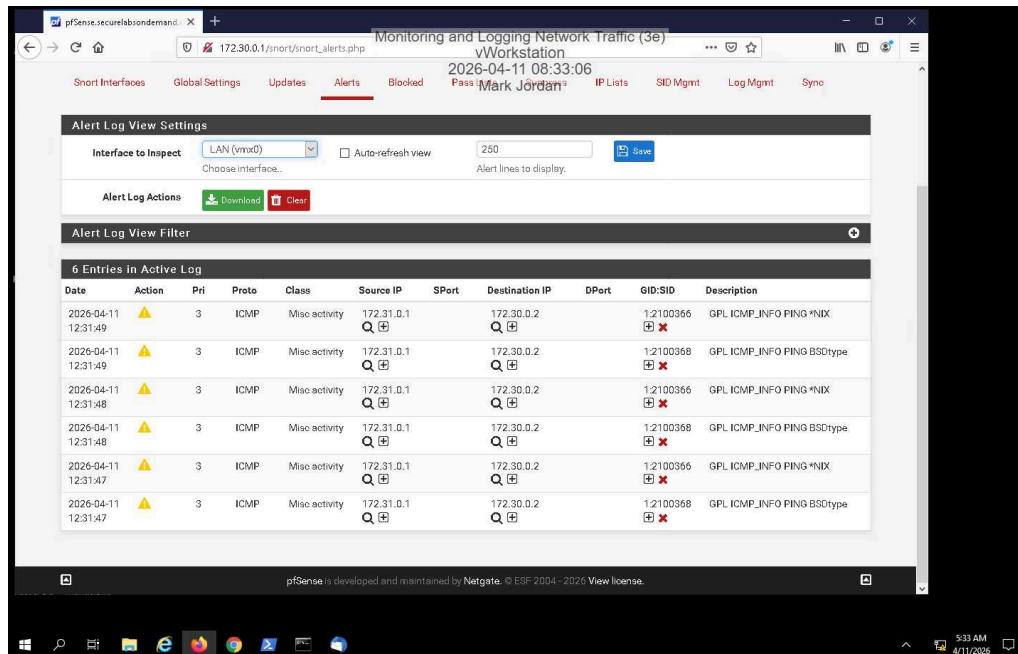
28. Make a screen capture showing the active Snort status on the LAN interface.



33. Make a screen capture showing the successful ping results.

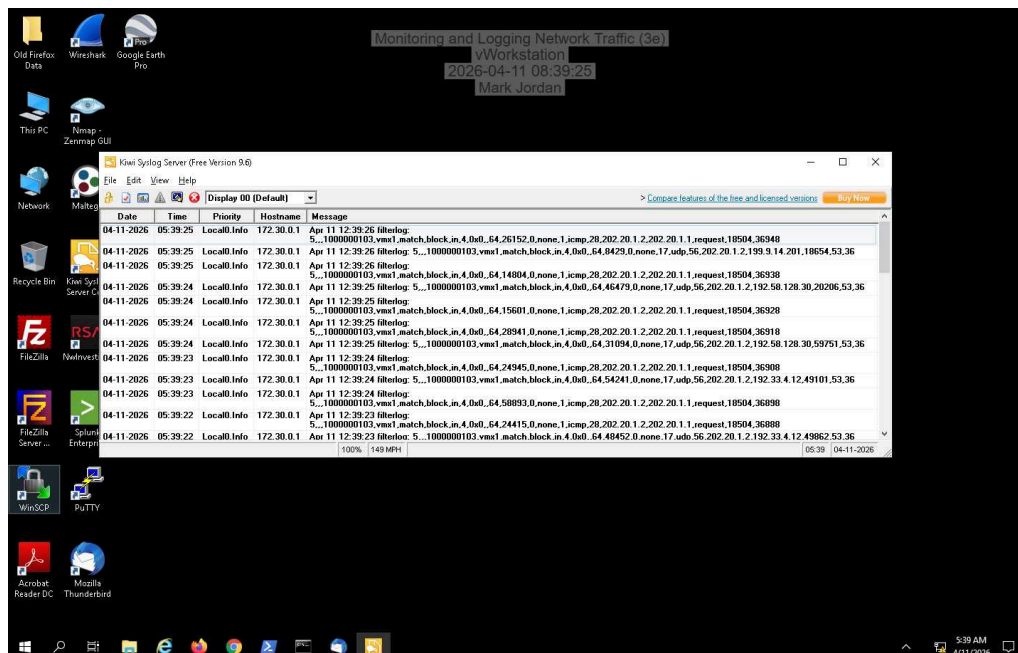


38. Make a screen capture showing the ICMP alerts in the Snort Active Log.



Part 3: Implement Firewall Log Forwarding with Kiwi Syslog Server

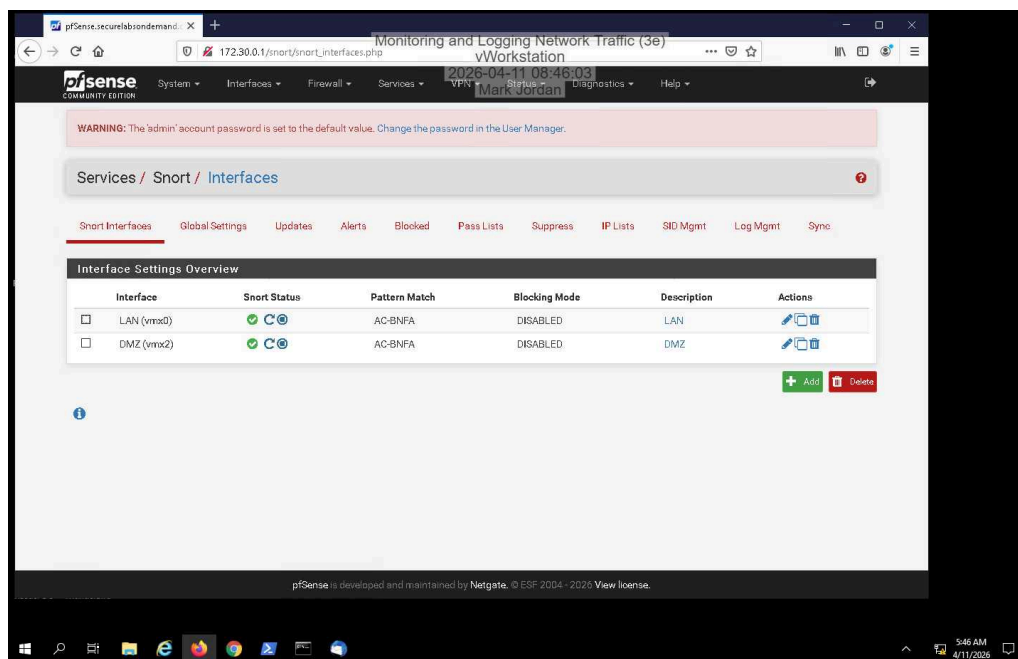
17. Make a screen capture showing the pfSense firewall log events in Kiwi Syslog Server.



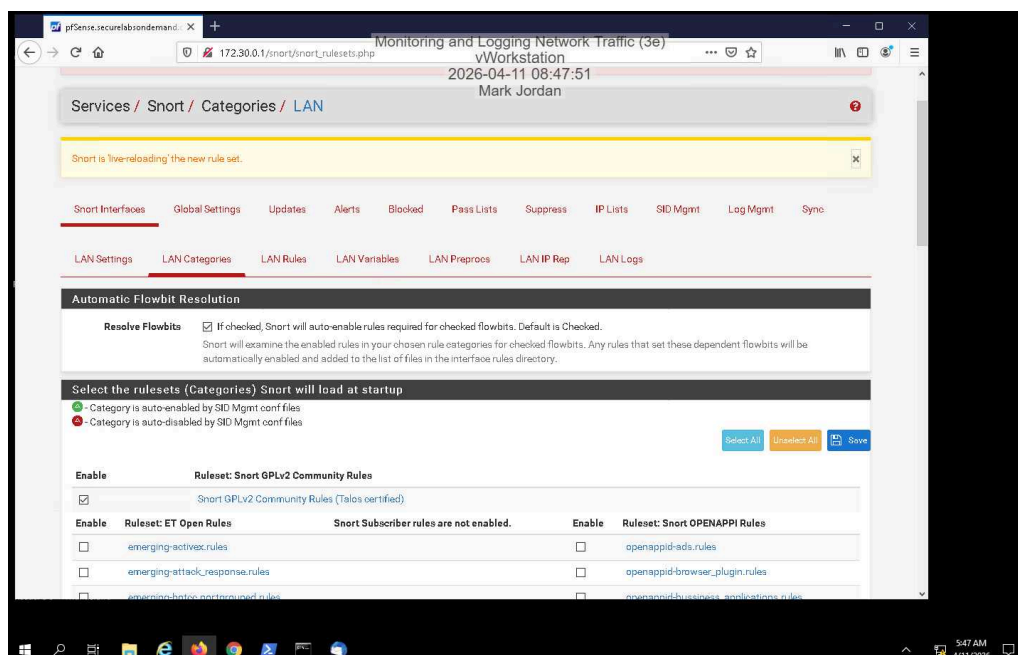
Section 2: Applied Learning

Part 1: Configure Snort Monitoring on the DMZ

17. Make a screen capture showing the active Snort status on the DMZ interface.

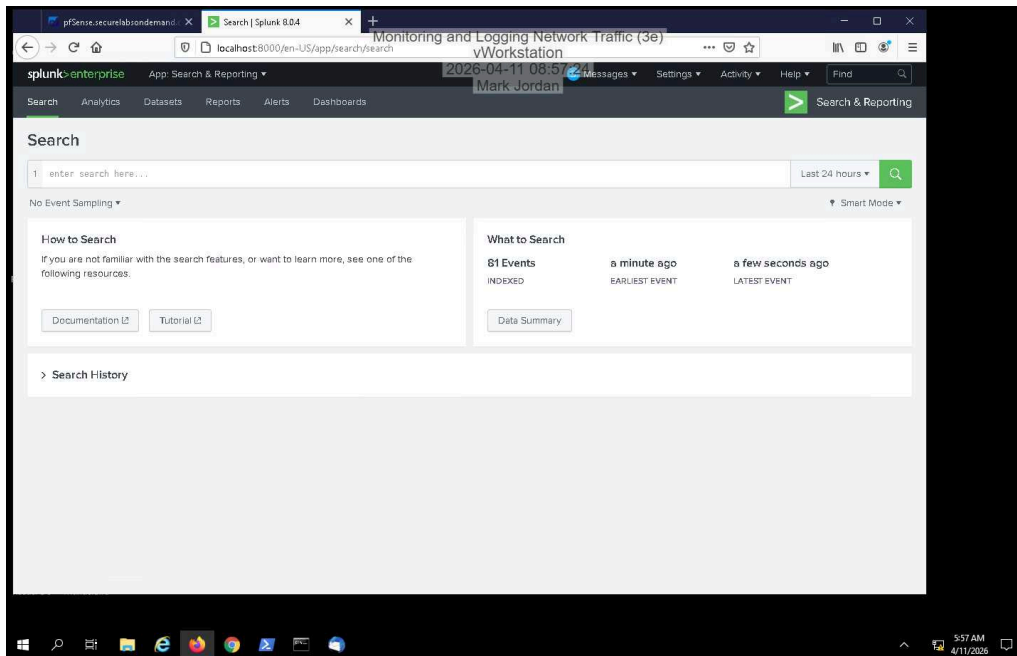


20. Make a screen capture showing the Snort GPLv2 Community Rules enabled and "live-reloading" message.



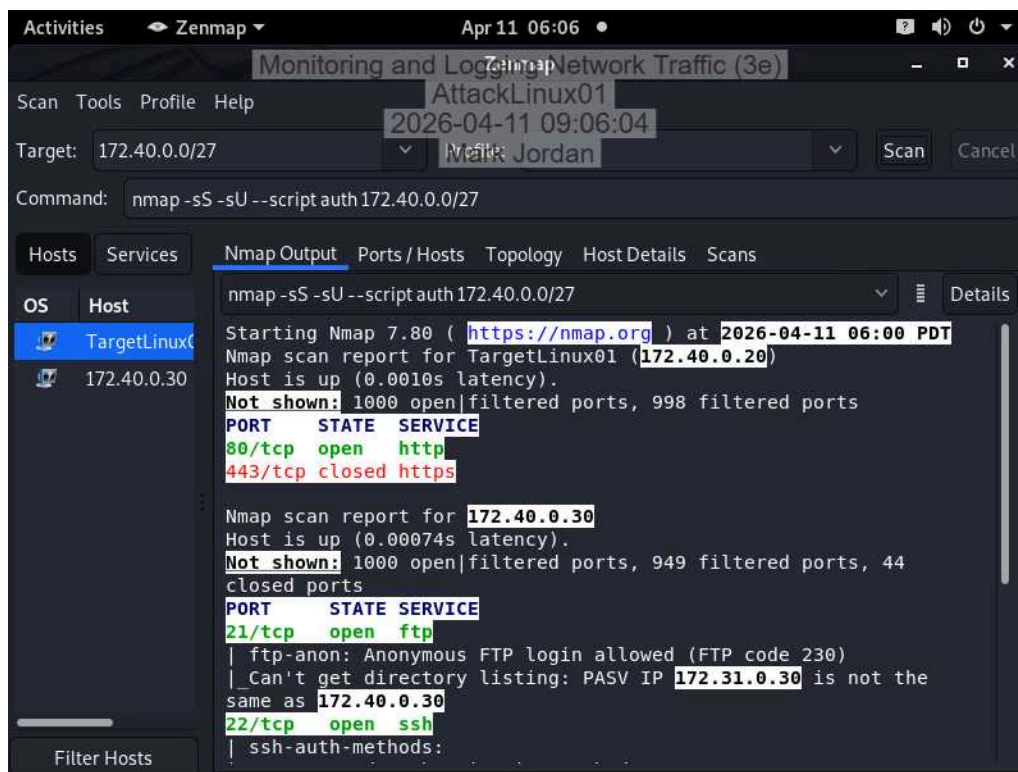
Part 2: Implement Security Information and Event Management with Splunk

13. Make a screen capture showing the indexed events in Splunk.

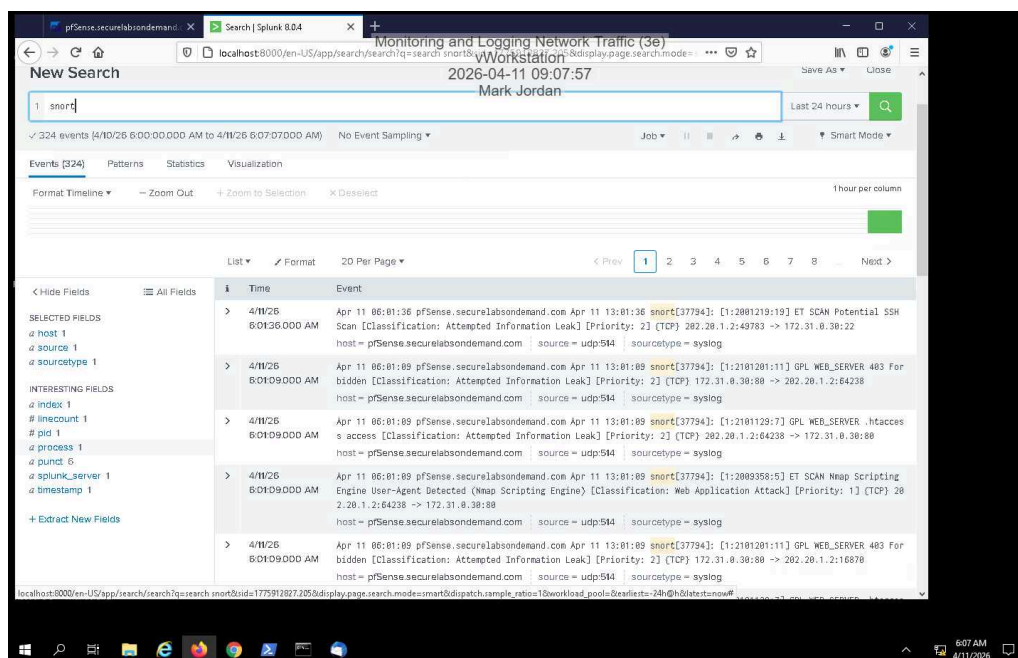


Part 3: Simulate and Detect a Perimeter Network Attack

6. Make a screen capture showing the Nmap scan report.



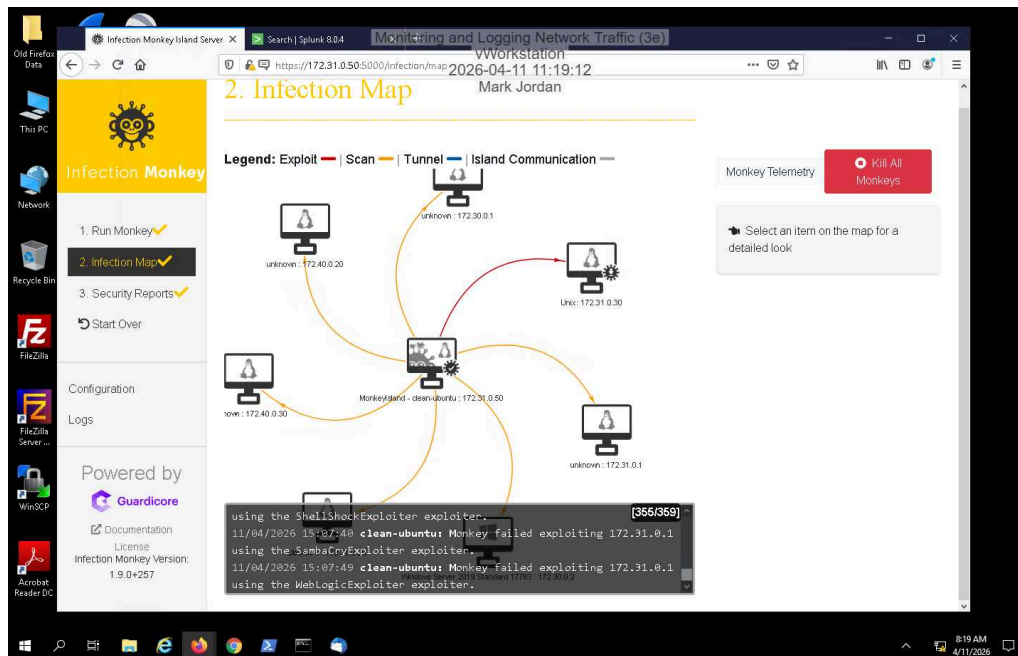
9. Make a screen capture showing the search results in Splunk.



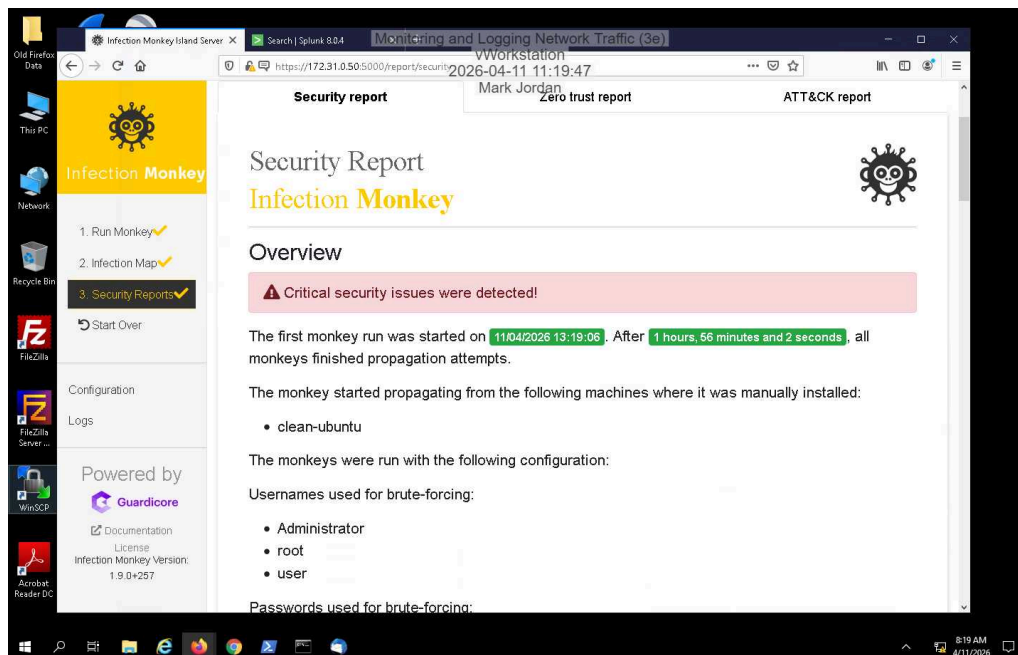
Section 3: Challenge and Analysis

Part 1: Simulate a DMZ Breach with Infection Monkey

Make a screen capture showing the resulting Infection Map.



Make a screen capture showing the resulting Security Report.

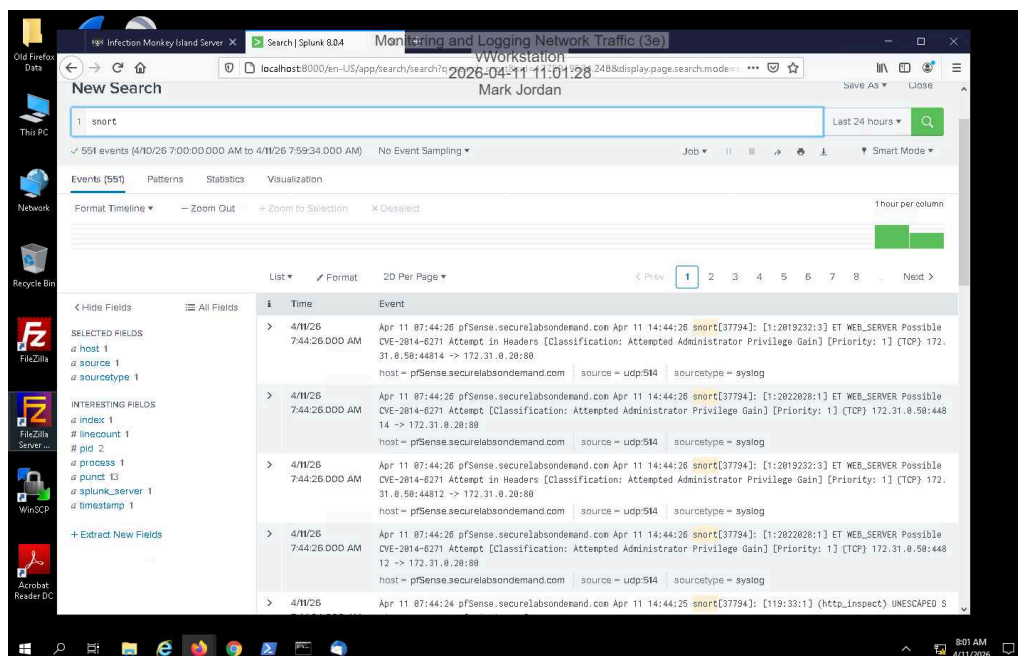


Summarize your DMZ breach simulation results, highlighting what you found to be the greatest concerns from a network monitoring perspective.

Based on my DMZ breach simulation results, the immediate threat to be of greatest concern is the vulnerable VSFTPD and weak segmentation. An outdated VSFTPD in the DMZ can act as an entry point for attackers. Along with the outdated VSFTPD vulnerability, weak segmentation would allow the attacker to move from the DMZ into more sensitive internal networks. It is recommended to update you VSFTPD server to a more secure version vsftpd-3.0.3 and to reconfigure firewalls to restrict network traffic communication between DMZ devices to only necessary protocols.

Part 2: Detect a Simulated DMZ Breach with Snort and Splunk

Make a screen capture showing the results of your search query for Infection Monkey traffic in Splunk.



Describe any concerns about the structure of the query result or the data elements it contains. What data fields would you add, remove, or edit to make log analysis more effective?

When searching "Snort" for Infection Monkey results, the Event details do provide a bit more legibility than just trying to search "172.31.0.50" (Infection Monkey IP address). The search "snort" method allows me to identify the IP address for Infection Monkey in the Events details along with the IP address that Infection Monkey is trying to attack. It also provides a description of the CVE and attempted exploit for said IP address. One data field I would add is "Request Duration/Response Time" to help identify slowloris attacks and performance bottlenecks.

Write a brief memo to your manager describing Splunk's usefulness in detecting traces of your simulated breach. What configuration changes would you recommend? How would you enhance its functionality?

Dear Manager,

Today's report describes Splunk's usefulness in detecting traces in my simulated breach. Splunk provides real-time dashboards for more visibility on anomalous activities such as lateral movements or data exfiltration by analyzing logs from firewalls and IDS's. Splunk is also able to detect failed log in attempts, unauthorized access attempts, and Infection Monkey's network traffic in real time. The configurations I would recommend are log forwarding optimization to ensure accurate log forwarding from the pfSense/Snort into Splunk and I would setup custom alert thresholds for specific indicators of compromise such as multiple failed login attempts, abnormal data transfers, and unauthorized access/lateral movements.